

9.BINARY SEARCH TREE (BST)

Description

A Java implementation of a Binary Search Tree supporting insertion, searching, deletion, and tree traversals.

Features

- Insert Node
- Search Node
- Delete Node
- Inorder Traversal
- Preorder Traversal
- Postorder Traversal

Technologies Used

- Java

Concepts Used

- Binary Search Tree
- Recursion
- Data Structures
- Tree Traversals

How to Run

Run `Main.java`.

Sample Output

```
===== Binary Search Tree =====
```

```
1. Insert Node
```

```
2. Search Node
```

```
3. Delete Node
```

```
4. Inorder Traversal
```

```
5. Preorder Traversal
```

```
6. Postorder Traversal
```

```
7. Exit
```

```
Enter your choice: 1
```

```
Enter value to insert: 50
```

```
Node inserted successfully!
```

```
===== Binary Search Tree =====
```

Enter your choice: 1

Enter value to insert: 30

Node inserted successfully!

===== Binary Search Tree =====

Enter your choice: 1

Enter value to insert: 70

Node inserted successfully!

===== Binary Search Tree =====

Enter your choice: 1

Enter value to insert: 20

Node inserted successfully!

===== Binary Search Tree =====

Enter your choice: 1

Enter value to insert: 40

Node inserted successfully!

===== Binary Search Tree =====

Enter your choice: 1

Enter value to insert: 60

Node inserted successfully!

===== Binary Search Tree =====

Enter your choice: 1

Enter value to insert: 80

Node inserted successfully!

===== Binary Search Tree =====

Enter your choice: 4

Inorder Traversal:

20 30 40 50 60 70 80

===== Binary Search Tree =====

Enter your choice: 5

Preorder Traversal:

50 30 20 40 70 60 80

===== Binary Search Tree =====

Enter your choice: 6

Postorder Traversal:

20 40 30 60 80 70 50

===== Binary Search Tree =====

Enter your choice: 2

Enter value to search: 60

Node found in Binary Search Tree.

===== Binary Search Tree =====

Enter your choice: 3

Enter value to delete: 30

Node deleted successfully!

===== Binary Search Tree =====

Enter your choice: 4

Inorder Traversal after deletion:

20 40 50 60 70 80

===== Binary Search Tree =====

Enter your choice: 7

Exiting Binary Search Tree Application...

Author

Gopavaram Suchithra.